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Siemens and Humanoid bring Physical AI to the factory floor: deploying humanoids in industrial operations with NVIDIA

- **Siemens' and NVIDIA's strategic partnership to build fully AI-driven, adaptive manufacturing sites delivers real-world milestone with testing of humanoid robot at Siemens factory in Erlangen, Germany**
- **HMND 01 Alpha is purpose-built for industrial environments with advanced manipulation capabilities**
- **Siemens Xcelerator offers a factory-grade scalable portfolio for industrial integration**
- **NVIDIA accelerates simulation and development for faster deployment**

Siemens and Humanoid today announced a landmark milestone in the journey to bring physical AI from vision to industrial reality. Humanoid's HMND 01 wheeled Alpha humanoid robot, built using the NVIDIA physical AI stack, has been successfully tested in operations at Siemens' electronics factory in Erlangen, Germany, performing autonomous logistics tasks. This builds on the Siemens and NVIDIA strategic partnership, announced at CES, to build the world's first fully AI-driven, adaptive manufacturing sites.

The dawn of physical AI in manufacturing

Physical AI - the discipline of training intelligent machines to perceive, reason and act in the physical world - is poised to transform how goods are made. Bridging the gap between AI research and the demands of a real factory requires a high-performing ecosystem: world-class AI compute and simulation, a proven robotics platform, and the deep industrial automation infrastructure to tie it all together.

The HMND 01 Alpha robot was deployed in Siemens' logistics operations, where it autonomously executed tote-handling tasks - picking, transporting and placing containers for human operators. All target performance metrics were met, including a throughput of 60 tote moves per hour, uptime exceeding 8 hours, and autonomous pick-and-place success rates above 90 percent.



Siemens and Humanoid Bring Physical AI to the Factory Floor: Deploying Humanoids in Industrial Operations with NVIDIA

Building the industrial backbone with Siemens Xcelerator

A humanoid robot's true value is in becoming a fully integrated, collaborative asset on the shop floor. That means real-time data exchange with production systems and other Autonomous Guided Vehicles, synchronized workflows with other machinery and human operators, and adaptive behavior that responds dynamically to changing conditions. Without this deep integration, even the most sophisticated robot remains an isolated feature.

Siemens provides this critical layer through its Siemens Xcelerator portfolio, from a comprehensive digital twin to AI-enabled perception, to integrated control and PLC-robot interfaces, along with fleet management, industrial communication networks and high-performance drives. Together, these technologies form the digital backbone and automation infrastructure that help to ensure humanoid robots

operate efficiently and in concert with the broader factory environment. The outcome is a factory-grade model for deploying humanoids in any industrial setting.

Accelerating intelligence with NVIDIA libraries, frameworks and AI infrastructure

Humanoid has integrated NVIDIA's full physical AI stack into the HMND 01 platform, including NVIDIA Jetson Thor for edge compute, NVIDIA Isaac Sim for simulation and NVIDIA Isaac Lab for reinforcement learning and policy training. The result is a dramatic compression of development timelines. Simulation-first hardware design has also enabled the team to optimize actuator selection, joint strength and mass distribution virtually, cutting prototype development from a typical 18–24 months to just 7 months.

"Factories of the future demand robots that can perceive, reason, and adapt autonomously alongside human workers, tackling the labor shortages and operational complexity that traditional automation struggled to handle," said Deepu Talla, vice president of Robotics and Edge AI at NVIDIA. "With Siemens providing the industrial integration backbone and Humanoid deploying NVIDIA's full physical AI stack - from simulation-first training to real-time edge inference - this deployment paves the way for humanoid robots meeting real production targets on a live factory floor."

Humanoid: Building factory-grade humanoids

Humanoid, a UK-based AI and robotics company, developed the HMND 01 Alpha, a humanoid robot purpose-built for industrial environments. Combining an omnidirectional wheeled mobility platform with advanced manipulation capabilities, powered by KinetIQ, a proprietary AI framework, the HMND 01 is engineered to work in human-centric spaces, adapting to diverse tasks and handling complex actions.

"Our mission is to create humanoid robots that perform not only in controlled lab settings, but also in real-world factory environments, handling meaningful industrial tasks. Our collaboration with Siemens and NVIDIA gives us a powerful advantage by combining NVIDIA's leading AI infrastructure, simulation tools, and frameworks with Siemens' deep industrial expertise and integration capabilities," said Artem

Sokolov, CEO and Founder of Humanoid. “Together, we’ve proven that humanoid robots are ready for real-world industrial deployment.”

This press release as well as press pictures are available at

<https://press.siemens.com/global/en/pressrelease/siemens-and-humanoid-bring-physical-ai-factory-floor-deploying-humanoids-industrial>

Further information on Siemens at Hannover Messe 2026 at

www.siemens.com/press/hm26 and www.siemens.com/hannover-messe

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Siemens Digital Industries (DI) empowers companies of all sizes within the process and discrete manufacturing industries to accelerate their digital and sustainability transformation across the entire value chain. Siemens' cutting-edge automation and software portfolio revolutionizes the design, realization and optimization of products and production. And with Siemens Xcelerator – the open digital business platform – this process is made even easier, faster, and scalable. Together with our partners and ecosystem, Siemens Digital Industries enables customers to become a sustainable Digital Enterprise. Siemens Digital Industries has a workforce of around 70,000 people worldwide.

Siemens AG (Berlin and Munich) is a leading technology company focused on industry, infrastructure, mobility, and healthcare. The company's purpose is to create technology to transform the everyday, for everyone. By combining the real and the digital worlds, Siemens empowers customers to accelerate their digital and sustainability transformations, making factories more efficient, cities more livable, and transportation more sustainable. A leader in industrial AI, Siemens leverages its deep domain know-how to apply AI – including generative AI – to real-world applications, making AI accessible and impactful for customers across diverse industries. Siemens also owns a majority stake in the publicly listed company Siemens Healthineers, a leading global medical technology provider pioneering breakthroughs in healthcare. For everyone. Everywhere. Sustainably.

In fiscal 2025, which ended on September 30, 2025, the Siemens Group generated revenue of €78.9 billion and net income of €10.4 billion. As of September 30, 2025, the company employed around 318,000 people worldwide on the basis of continuing operations. Further information is available on the Internet at www.siemens.com.

Humanoid is a UK-based robotics innovation company dedicated to developing advanced humanoid robots that enhance human capabilities. Founded by Artem Sokolov in 2024, Humanoid brings together over 200 engineers, researchers, and innovators from top global tech companies. With offices in London, Boston, and Vancouver, the company is focused on building commercially viable, scalable, and safe robotic solutions for real-world applications.